基于LADRC的纯电动汽车双离合变速器换挡控制㊀㊁

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【摘要】 针对纯电动汽车两挡双离合自动变速器（two-speed dual-clutch transmission，2DCT）换挡控制模型及参数的不确定性和存在的未知干扰，提出了一种线性自抗扰（linear active disturbance rejection controller，LADRC）换挡控制策略。首先建立2DCT换挡过程动力学模型，并分析了换挡过程；然后考虑控制模型的不确定性及未知干扰，将LADRC控制器应用到换挡过程对期望角速度进行跟踪，并采用扩张状态观测器（extended state observer，ESO）对扰动进行实时估计，并加以补偿，最后与PID控制进行对比。仿真和实验结果表明，该控制器跟踪误差更小，鲁棒性强，能保证良好的换挡品质。

【关键词】 纯电动汽车，双离合变速器，换挡控制，自抗扰控制，扩张状态观测器

Control Strategy for Dual-Clutch Transmission Gear Shifting Process in Pure Electric Vehicles Based on LADRC

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Abstract：For the uncertainty of the shift control model and parameters and the existing unknown distur-bances of the two-speed dual clutch transmission（2DCT）of pure electric vehicles，a linear active disturbance rejec-tion controller（LADRC）method is proposed. Firstly，the dynamic model of the 2DCT shifting process is estab-lished，and the shifting process is analyzed. Then，considering the uncertainty and unknown disturbance of the con-trol model，the LADRC controller is applied to the shifting process to track the desired rotational speed，and the Ex-tended State Observer（ESO）is used to estimate the disturbance in real time and compensate for it. Finally，it is compared with the PID control. Simulation and experimental results show that the controller has smaller tracking er-ror and strong robustness，which ensures good gear shifting quality.

Key words：pure electric vehicles，dual-clutch transmissions，shifting control，active disturbance rejection controller，extended state observer

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